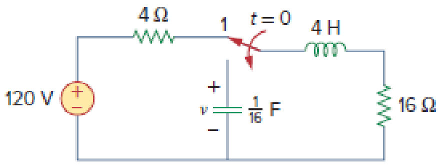
Chapter 16, Problem 54P

Problem

The switch in Fig. 16.77 has been in position 1 for $t < 0$. At $t = 0$, it is moved from position 1 to the top of the capacitor at $t = 0$. Please note that the switch is a make before break switch; it stays in contact with position 1 until it makes contact with the top of the capacitor and then breaks the contact at position 1. Determine $v(t)$.

Figure 16.77:



Step-by-step solution

Step 1 of 5

Refer to circuit diagram in Figure 16.77 in the textbook.

The switch has been in position 1 for $t < 0$. Thus, the circuit will remain in the steady state in position 1 and the inductor connected will be replaced by a short circuit.

Redraw the circuit as shown in Figure 1.

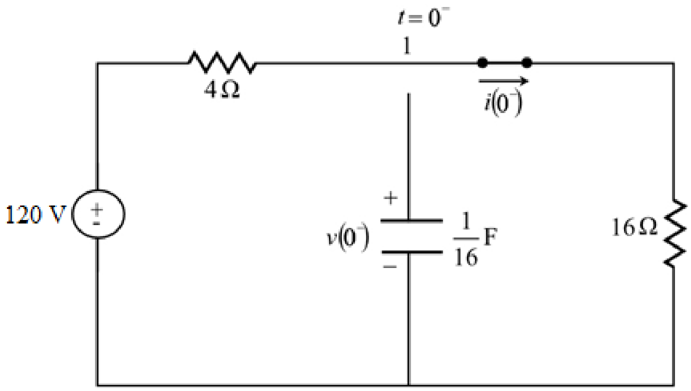


Figure 1

Step 2 of 5

Apply Kirchhoff's voltage law in Figure 1.

$$4i(0^-)+16i(0^-)=120$$
$$20i(0^-)=120$$
$$i(0^-)=\frac{120}{20}$$
$$=6\text{ A}$$

Since the capacitor is disconnected from the circuit, the voltage across the capacitor at $t=0^-$ is zero, that is,

$$v(0^-)=0\text{ V}$$

Step 3 of 5

At $t=0$, the switch is connected to the capacitor branch. Replace the circuit by its s-domain equivalent for this connection as shown in Figure 2.

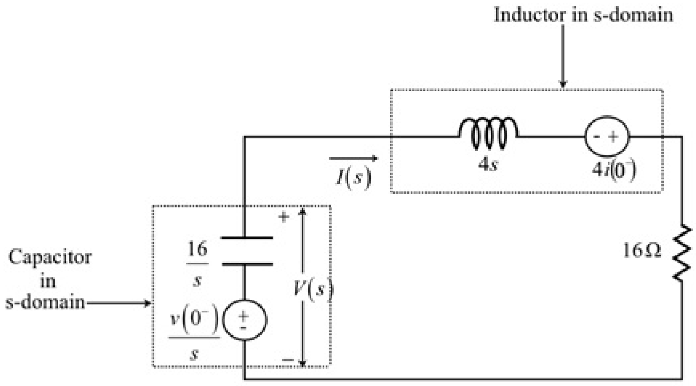


Figure 2

Step 4 of 5

Apply Kirchhoff's voltage law in Figure 2.

$$-\frac{v(0^-)}{s}+\left(\frac{16}{s}+4s+16\right)I(s)-4i(0^-)=0$$

Substitute $6\,\mathrm{A}$ for $i(0^-)$ and $0\,\mathrm{V}$ for $v(0^-)$.

$$-\frac{0}{s}+\left(\frac{16}{s}+4s+16\right)I(s)-(4)(6)=0$$

$$\left(\frac{16+4s^2+16s}{s}\right)I(s)=24$$

$$I(s)=\frac{6s}{s^2+4s+4}$$

Step 5 of 5

Calculate $V(s)$ from Figure 2.

$$V(s)=\left(\frac{16}{s}\right)\left[-I(s)\right]-\frac{v(0^-)}{s}$$

Substitute $\frac{6s}{s^2+4s+4}$ for $I(s)$ and $0\,\mathrm{V}$ for $v(0^-)$.

$$\begin{aligned}V(s)&=\left(\frac{16}{s}\right)\left(-\frac{6s}{s^2+4s+4}\right)-0\\&=-\left(\frac{16}{s}\right)\left(\frac{6s}{(s+2)^2}\right)\\&=-\frac{96}{(s+2)^2}\end{aligned}$$

Apply inverse Laplace transform.

$$v(t)=-96te^{-2t}u(t)\ \mathrm{V}$$

Therefore, the value of $v(t)$ is $\boxed{-96te^{-2t}u(t)\,\mathrm{V}}$.